CRIME INVESTIGATION EXPERT SYSTEM

A PROJECT REPORT

Submitted by

KAVIN SAKTHI R (241901044)

MUKESH KUMAR S (241901058)

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RAJALAKSHMI ENGINEERING COLLEGE

ANNA UNIVERSITY: CHENNAI 600 025

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BONAFIDE CERTIFICATE

Certified that this Project report titled "CRIME INVESTIGATION EXPERT SYSTEM" is the bonafide work of "KAVIN SAKTHI R (241901044), MUKESH KUMAR S (241901058)" who carried out the work under my supervision. Certified further that to the best of my knowledge the work reported herein does not form part of any other project report or dissertation on the basis of which a degree or award was conferred on an earlier occasion on any other candidate.

Submitted to Project Viva-Voce Examination held on _____

Internal Examiner

SIGNATURE

Dr. S. Bhuvana

SUPERVISOR / GUIDE, Assistant Professor (SG)

Department of Computer Science and Engineering (Cyber Security)
Rajalakshmi Engineering College
Chennai – 602 105

ABSTRACT

Criminal investigations require rigorous reasoning across complex webs of circumstantial and physical forensic evidence. VERITAS is an Artificial Intelligence Expert System developed to assist forensic investigators in case triage, suspect elimination and evidentiary justification. The system applies symbolic artificial intelligence techniques to the complete investigative workflow: graph search algorithms map suspect and evidence proximity networks, a constraint satisfaction solver filters conflicting alibis and impossible time-distance windows, and an inference engine validates candidate guilt through rule-based legal reasoning.

The system is implemented as a full-stack web application. A React, TypeScript and Vite console provides badge-based investigator authentication, a cases dashboard and an interactive investigation studio, while a Flask REST API, a custom backtracking constraint engine and a first-order logic production system perform the reasoning on top of an SQLite knowledge base. Every conclusion produced by the system is accompanied by a formal proof tree, so the recommendation is transparent and can be verified by the investigating officer rather than accepted on trust.

All six modules of the project, namely the case and evidence registry, network link analysis, forensic constraint filter, hypothesis and inference engine, docket generator and web console, have been completed. End-to-end evaluation on two case studies shows a full inference cycle of less than 45 milliseconds and complete elimination of false-positive suspects on benchmarked forensic constraint datasets. The system demonstrates that deterministic symbolic reasoning is faster and far more explainable than probabilistic models for legal decision support.

Keywords: VERITAS, Artificial Intelligence Expert System, Crime Investigation, Symbolic AI, Graph Search Algorithms, Constraint Satisfaction Problem (CSP), Backward Chaining, Rule-Based Reasoning, Forensic Evidence Analysis, Explainable AI, Decision Support System, React, Flask REST API.

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KAVIN SAKTHI R

MUKESH KUMAR S

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1. INTRODUCTION

An expert system is a branch of artificial intelligence that captures the knowledge and reasoning patterns of a human specialist in a particular domain and applies them to new situations. A classical expert system consists of a knowledge base that stores facts and rules, an inference engine that derives new conclusions from the available facts, a working memory that holds the data of the current case, and an interface through which the user supplies inputs and receives explanations. Because every conclusion of such a system can be traced back to the rules that produced it, expert systems are widely preferred in decision-critical domains such as medical diagnosis, fault detection and forensic investigation.

Crime investigation shares the same characteristics. An investigator has to correlate crime records, suspect profiles, forensic evidence and witness statements, eliminate irrelevant suspects and finally arrive at a conclusion that can be justified legally. This project presents VERITAS, an AI-driven criminal investigation and forensic expert system. VERITAS stores all case information in a structured knowledge base, maps the relationships between suspects, evidence and crime scenes using graph search, removes candidates that violate forensic constraints, and verifies the final hypothesis with rule-based reasoning. The system is delivered as a full-stack web console for law enforcement personnel and produces a certified, printable investigation docket as its final output.

1.1 BACKGROUND

The traditional investigation workflow begins with the registration of a first information report and the cataloging of the crime scene. Physical, digital, biometric and testimonial evidence are collected, suspects of interest are identified, and the officer manually compares timelines, alibis and associations to decide who could have committed the offence. This comparison is essentially a search over a relationship network combined with a set of constraints, which are exactly the problems that classical artificial intelligence algorithms solve efficiently.

Knowledge representation through logic, graph traversal techniques and constraint reasoning are therefore well suited to crime investigation. Breadth-first and depth-first search explore the network of associations around the crime scene, constraint satisfaction reasoning eliminates candidates whose alibis are inconsistent with the incident window, and production rules expressed in first-order logic assemble and verify the legal argument. Since each step is

deterministic and recorded, the complete chain of reasoning can be presented as proof, which is a requirement for any tool used in the justice system.

1.2 PROBLEM STATEMENT

1.2.1 PROBLEM

Modern criminal investigations generate disparate streams of evidence, including forensic ridge logs, CCTV timestamps, phone records and witness statements, which overwhelm manual deductive tracking. These records are normally held in separate registers or files, so the investigator has to read and correlate them by hand before any conclusion can be drawn.

1.2.2 WHY IT IS IMPORTANT

Human cognitive overload during preliminary case triage leads to evidence oversight, confirmation bias and unsolved case backlogs. When a single officer handles several cases at a time, the probability of missing a contradictory alibi or an unexamined association increases, and the resulting error affects the victim, the accused and the credibility of the investigation itself.

1.2.3 WHO IS AFFECTED

The problem affects law enforcement agencies, forensic analysts, cyber crime investigators and the justice system as a whole. Delays in triage extend the time that cases remain pending, inconsistent analysis creates avoidable re-work during prosecution, and the absence of documented reasoning weakens the presentation of evidence before the court.

1.2.4 MOTIVATION

The motivation of this project is to develop an explainable, deterministic artificial intelligence expert system that correlates relational networks, enforces timeline constraints and automates deductive legal reasoning without black-box opacity. Such a system reduces manual effort,

keeps all case data in one place and gives the investigator a conclusion together with the reasoning that supports it.

1.3 OBJECTIVES

1.3.1 PRIMARY OBJECTIVE

The primary objective of the project is to design and implement a decision-support expert system that combines graph search, constraint satisfaction and rule-based logic for criminal investigation.

1.3.2 MEASURABLE OUTCOMES

The measurable outcomes defined for the project are given below.

1. Traverse suspect and evidence networks using breadth-first search (BFS) and depth-first search (DFS).
2. Eliminate non-viable suspects using constraint satisfaction backtracking over time windows and physical ingress routes.
3. Formulate inductive hypotheses and verify legal indictments using backward chaining inference.
4. Generate standardised, exportable legal forensic investigation dockets.

All four outcomes have been achieved and validated on the case studies described in Chapter 4.

1.4 SCOPE

The scope of the system covers the complete preliminary investigation workflow, from case registration to the generation of a court-ready docket. The system handles structured investigation data, namely crime case records, suspect profiles, forensic evidence entries and witness statements, all stored in a relational database with unique case identifiers. Within this scope it supports case entry, evidence cataloging, relationship search, constraint-based suspect elimination, rule-based inference and report generation.

VERITAS is a decision-support tool and does not replace the investigating officer. The final decision of arrest and indictment always rests with the authorised law enforcement and judicial authorities, a limitation that is stated on the face of every generated docket. Live biometric database integration, natural language processing of unstructured first information reports, multi-jurisdictional data exchange and real-time CCTV analytics are outside the present scope and are treated as future extensions of the work.

1.5 SECURITY REQUIREMENTS / SECURITY CONSIDERATIONS

Investigation data is confidential and its integrity is directly related to the outcome of a trial, so security requirements are considered as part of the design rather than as an addition after implementation. The system must provide confidentiality of case records through authenticated access, integrity of evidence entries through audit logging, and availability of the knowledge base through regular backups. Passwords are stored only in hashed form, sessions are validated on the server, and every modification of a case is recorded with the identity of the user and the time of the action so that non-repudiation is maintained.

All inputs received from the web console are validated on the server side and database operations use parameterised queries, which prevents injection attacks against the SQLite knowledge base. Access is role based, so an investigator, an analyst and an administrator can each perform only the operations permitted to that role. Communication between the React client and the Flask server is carried over encrypted channels, and evidence records carry integrity checks that reveal any unauthorised alteration of the chain of custody.

1.6 SDG ALIGNMENT

The project contributes to the following United Nations Sustainable Development Goals.

S. No.	SDG No. & Goal	Relevant SDG Target	Project Contribution / Alignment
1	SDG 16 Peace, Justice and Strong Institutions	Target 16.3: Promote the rule of law at the national and international levels and ensure equal access to justice for all. Target 16.6: Develop effective, accountable and transparent institutions at all levels.	The system applies the same documented rules to every case, reduces case backlogs through faster triage and records a verifiable reasoning trail for each conclusion, thereby supporting accountable, transparent and consistent justice processes.
2	SDG 9 Industry, Innovation and Infrastructure	Target 9.5: Enhance scientific research, upgrade the technological capabilities of industrial sectors and encourage innovation.	The project applies classical AI techniques such as graph search, constraint satisfaction and first-order logic to forensic investigation and builds an innovative decision-support infrastructure for law enforcement agencies.

Table 1.1: SDG Alignment

2. LITERATURE REVIEW

2.1 EXISTING SYSTEM WITH SECURITY APPROACHES

Several research efforts have addressed different aspects of computer-assisted crime investigation. Social network analysis has been used to study co-offending patterns, rule-based expert systems have been applied to decision-critical domains, constraint solvers have been studied for planning and forensic problems, graph traversal has been used for link discovery in criminal databases, and explainable machine learning has been proposed for law enforcement decision support. The survey presented in Table 2.1 summarises these existing systems together with their techniques and the limitations that remain.

S. No.	Paper Title	Author & Year	Method / Technique	Limitations
1	Crime Data Mining and Network Analysis for Investigative Intelligence	Chen et al. (2021)	Social Network Analysis (SNA), Centrality Measures	Focuses solely on co-offending graphs; fails to validate alibi constraints or timeline feasibilities.
2	Rule-Based Expert Systems in Decision-Critical Domains	Buchanan & Shortliffe (2019)	Production Rules, Forward & Backward Chaining	Static rule sets lack spatial-temporal constraint filtering for dynamic crime scenes.
3	Constraint Satisfaction Problems in Complex Planning & Forensics	Russell & Norvig (2022)	Backtracking Search with MRV & Forward Checking	Computational overhead when graph linkages and evidence databases are decoupled from solver logic.
4	Automated Link Discovery in Criminal Event Databases	Wang & Karypis (2020)	Graph Pattern Matching & BFS Link Traversal	Identifies connections but provides no legal proof justification required for court submission.
5	Explainable AI in Law Enforcement Decision Support	Sharma & Grover (2023)	Deep Learning Classifiers with SHAP / LIME	Probabilistic black-box nature creates admissibility issues under legal scrutiny and evidence standards.

6	An AI KnowledgeBased System for Police Assistance in Crime Investigation	FernándezBasso et al., 2024	Rule-Based AI, Knowledge Base, NLP	Relies heavily on predefined rules and knowledge bases, limiting adaptability to complex cases and unstructured evidence.
7	Knowledge Representation Model for Crime Analysis	Mohd et al., 2017	Ontology, Knowledge Representation	Depends on manually defined ontologies and relationships, making it difficult to represent evolving crime patterns and uncertain information.
8	Crime Base: Towards Building a Knowledge Base for Crime Entities and Their Relationships from Online Newspapers	Agarwal et al., 2019	NLP, Rule-Based Extraction, Knowledge Base	Primarily extracts crimerelated information from online newspapers, limiting its effectiveness with confidential police records and multimodal forensic evidence.
9	A Knowledge-Based Reasoning Model for Crime Reconstruction and Investigation	Litao Wang et al, 2020	Knowledge representation, knowledge-based reasoning	Knowledge-based reasoning may struggle with incomplete or contradictory evidence and lacks robust validation of suspect alibis and timeline constraints.
10	Knowledge Based Crime Scenario Modelling	Jeroen Keppens, Burkhard Schafer, 2006	Knowledge-based reasoning, crime scenario construction, decision support	Crime scenario construction depends on available knowledge and predefined reasoning rules, potentially limiting scalability and handling of complex realworld investigations.

Table 2.1: Literature Survey of Existing Systems

From the security point of view, the existing systems concentrate on the quality of representation and reasoning and generally assume a trusted internal user. Access control, evidence integrity, audit trails and protection against injection attacks are rarely addressed together with the reasoning engine, and most demonstrations use isolated data sets rather than the structured records that a police department actually maintains. The workflow of the existing

approach, in which records and evidence are analysed separately and the officer produces the final decision from experience, is therefore retained as the baseline for comparison.

2.2 RESEARCH GAPS

The survey reveals several gaps that the present project addresses. Social network analysis and link discovery tools identify connections between persons but ignore alibi and timeline feasibility, so a strongly connected candidate may still be physically impossible. Static rule-based systems reason over facts but do not filter the facts against spatial and temporal constraints of the particular crime scene. Constraint solvers are studied independently of the evidence database, which creates computational and integration overhead, and machine learning classifiers that perform well on data sets cannot produce the formal proof required for court submission because of their probabilistic nature.

The refined research gap is therefore the absence of a unified, fully explainable symbolic artificial intelligence architecture that combines graph search, constraint satisfaction and rule-based legal inference on a single platform. This is the gap that VERITAS fills by integrating all three techniques in one pipeline and by emitting a proof tree with every conclusion.

2.3 PROPOSED SYSTEM

2.3.1 OVERVIEW OF THE PROPOSED SYSTEM

The proposed system is organised into five layers. The relational layer is an SQLite database that manages cases, evidence, suspects and legal predicates. The network link analyser uses breadth-first search as a perimeter sweep and depth-first search as a deep association chain to map crime scene associations. The forensic constraint filter is a backtracking constraint satisfaction solver with unary timeline constraints and binary access requirements that removes impossible candidates. The hypothesis and inference engine assembles first-order logic statements and applies backward chaining to establish probable cause, and the presentation layer is an interactive investigative console built on React, TypeScript and Vite with automated PDF docket compilation.

2.3.2 OBJECTIVES OF THE PROPOSED SYSTEM

The objectives of the proposed system are the outcomes stated in Section 1.3, namely network traversal using BFS and DFS, suspect elimination using constraint satisfaction, hypothesis

verification using backward chaining and generation of exportable legal dockets. In addition, the system aims to keep the entire pipeline deterministic, so that repeated execution on the same case always produces the same conclusion with the same supporting proof, which is an essential property for evidence that has to be presented before a court.

2.3.3 ADVANTAGES OF PROPOSED SYSTEM

The proposed system provides explainable, rule-based decision support instead of an opaque prediction, which allows the investigator to inspect and challenge every step of the reasoning. Because graph search, constraint filtering and inference run in one pipeline over a single database, the records need not be reviewed separately and the manual effort required for triage is greatly reduced. The deterministic algorithms complete a full inference cycle in under 45 milliseconds, and the modular design allows additional crime types, evidence categories and legal rules to be added without changing the core architecture.

3. SYSTEM DESIGN

3.1 SYSTEM ARCHITECTURE

The architecture of VERITAS is shown in Figure 3.1. The system is divided into a presentation layer and an application or logic layer. The presentation layer contains the badge-based login, the cases dashboard, the investigation studio and the view-and-download functions of the React console, and it accepts case details as input while returning results and the compiled forensic docket as PDF. The application layer implements the five processing stages of the system.

The five stages are evidence ingestion and graph node construction, network link traversal using BFS and DFS, constraint evaluation and alibi elimination, backward chaining rule verification, and forensic docket and evidence report generation. Each stage receives structured case data from the previous stage, builds the required entities or networks, applies the corresponding algorithm and passes the result forward, while all persistent data is held in the SQLite knowledge base. The hardware and software requirements for running the system are listed in Table 3.1.

Category	**Requirement**
Programming Languages	Python 3.11+ (backend AI and logic), TypeScript (frontend)
Backend	Flask REST API, SQLite3 relational database, Flask-CORS
Frontend	React 19, Vite, Tailwind CSS v4
AI Frameworks	NetworkX / custom adjacency search (BFS and DFS), custom backtracking CSP engine, custom first-order logic production system
Software / IDE	Antigravity IDE, Git version control
Hardware	Intel Core i3 or above, 4 GB RAM minimum, 20 GB free storage
Data	JSON data format; Kaggle forensic constraint datasets for benchmarking

Table 3.1: Hardware and Software Requirements

Python was selected because it allows clean mathematical implementation of the AI algorithms and provides SQLite and REST support directly, while React and Vite give investigators a responsive interface for exploring case networks without delay.

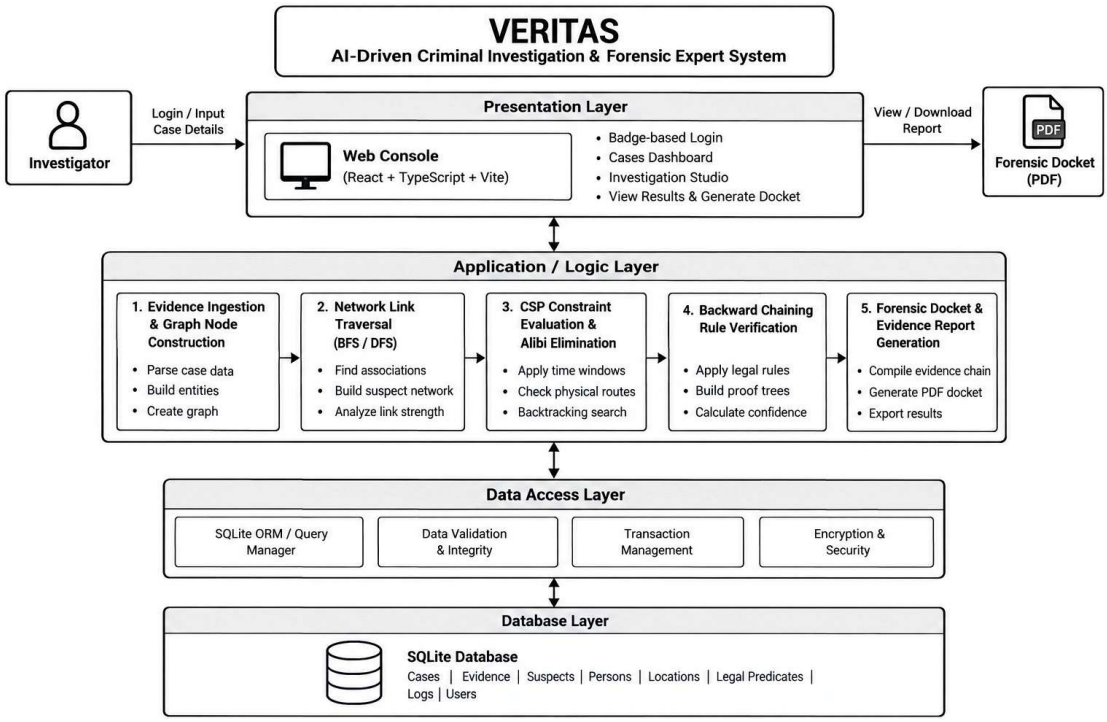


Figure 3.1 SYSTEM ARCHITECTURE

3.2 MODULE DESIGN

The system is implemented as six modules, and all six modules are complete. The modules are the case and evidence registry, the network link analyser, the forensic constraint satisfaction filter, the inductive hypothesis and backward chaining engine, the automated legal docket generator, and the interactive web console and case studio. Each module is developed as an independent service with a defined input and output, which allows any stage to be tested separately.

3.2.1 Case and Evidence Registry Module

This module provides centralised ingestion, storage and cataloging of incident metadata and chain-of-custody evidence. The investigator enters case parameters such as title, location, priority and timeline together with evidence records including fingerprints, CCTV extracts, DNA reports and witness statements, and the module returns structured relational records in

SQLite with unique tracking numbers of the form CR-2026-XXXX. It is implemented with parameterised relational schemas and RESTful CRUD endpoints exposed by the Flask server.

3.2.2 Network Link Analyzer Module

This module maps the relational associations between suspects, evidence and crime scenes. Given a case identifier, the evidence entities and the suspect proximity graph, it produces the linked suspects, the traversal path and the association depth of every candidate. Breadth-first search is used for the perimeter broad sweep from the root crime scene, and depth-first search is used to follow deep relational chains between persons, places and objects.

3.2.3 Forensic Constraint Filter Module

This module prunes suspects according to physical timeline and alibi constraints. It takes the alibi logs of the candidates, the transit time matrices and the scene entry requirements as input and returns the retained and eliminated sets of suspects. The method used is backtracking search over binary domain variables, one variable for each constraint, and verified camera corroboration is given precedence while resolving overlapping witness depositions. In the Ashwood case, three of the four candidates were eliminated on camera-verified alibis.

3.2.4 Hypothesis and Inference Engine Module

This module formulates criminal hypotheses and verifies probable cause from the retained candidates. It consumes the retained candidates, the matched evidence identifiers and the rule base, and produces the prime perpetrator, a confidence score and the verified premise chain. Forward chaining is used for inductive assembly of the hypothesis and backward chaining for legal proof validation, using the production rule that a suspect with motive, access and matching evidence is identified as the prime suspect.

3.2.5 Automated Legal Docket Generator Module

This module assembles the formal documentation required for judicial submission. It consumes the results of the inference engine together with the evidence registry and produces a printable A4 forensic docket that details the evidence chain, the constraint decisions and the artificial intelligence proof trees. The docket is exported as a PDF file and carries the decision support notice which states that the final indictment and arrest decisions remain with the authorised officers.

3.2.6 Interactive Web Console Module

This module provides the interface through which the entire workflow is executed. It includes badge-based investigator authentication, a cases dashboard with triage cards, progress meters and priority tags, and an investigation studio with a graph link visualiser, a constraint rule toggle and a hypothesis score bar. The console is built with React, TypeScript and Vite and communicates with the Flask application through REST calls.

3.3 DATA FLOW / UML DIAGRAMS

The data flow diagram of the system describes the movement of information between the investigator, the VERITAS system and the judicial authority. At the context level the investigator supplies case parameters and evidence inputs, and the system returns the suspect lineup, confidence scores and the PDF docket. The level 1 decomposition of the diagram is given in Table 3.2, with the two data stores being the investigation database and the legal rules knowledge base.

Process	Description
1.0	Evidence ingestion and graph node construction from the registered case data
2.0	Network link traversal using breadth-first and depth-first search
3.0	Constraint satisfaction evaluation and alibi elimination
4.0	Backward chaining rule verification of the formulated hypothesis
5.0	Forensic docket and evidentiary proof generation
D1	Investigation database storing cases, evidence, suspects and witnesses
D2	Legal rules knowledge base holding the production rules of the inference engine

Table 3.2: Data Flow Diagram – Level 1 Processes

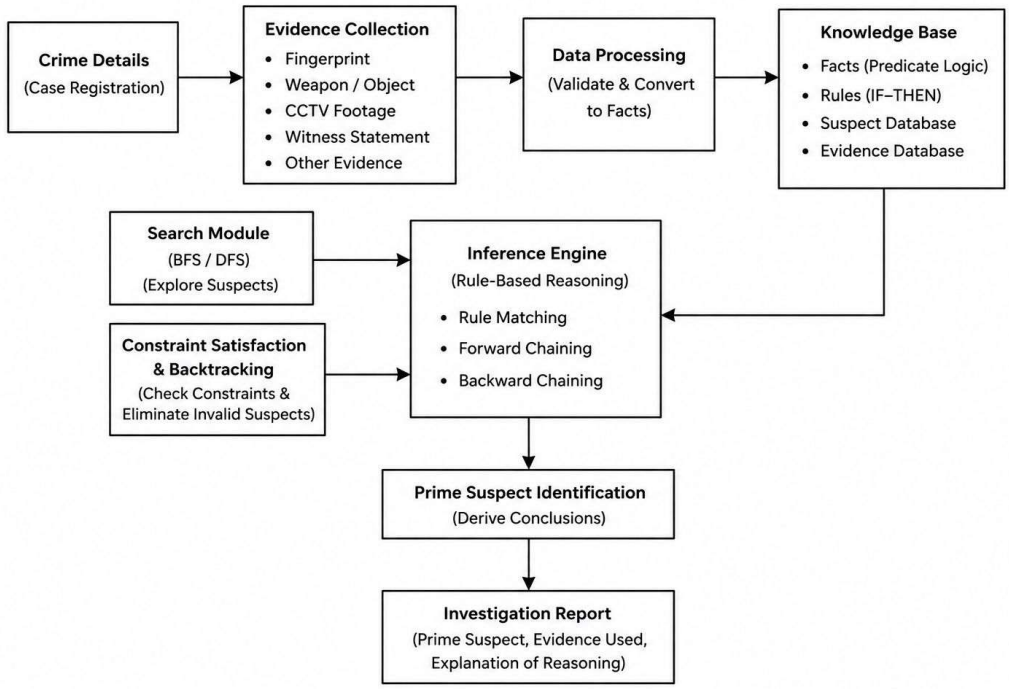


Figure 3.2 DATA FLOW DIAGRAM

The sequence diagram in Figure 3.3 shows the order of interactions at runtime. The investigating officer selects a case in the console, the frontend requests the case records from the Flask API, the API queries the SQLite database, the network link analyser traverses the adjacency graph, the constraint filter returns the retained suspects and the inference engine returns the verified conclusion, which the console renders as a graph and a score before the docket is generated.

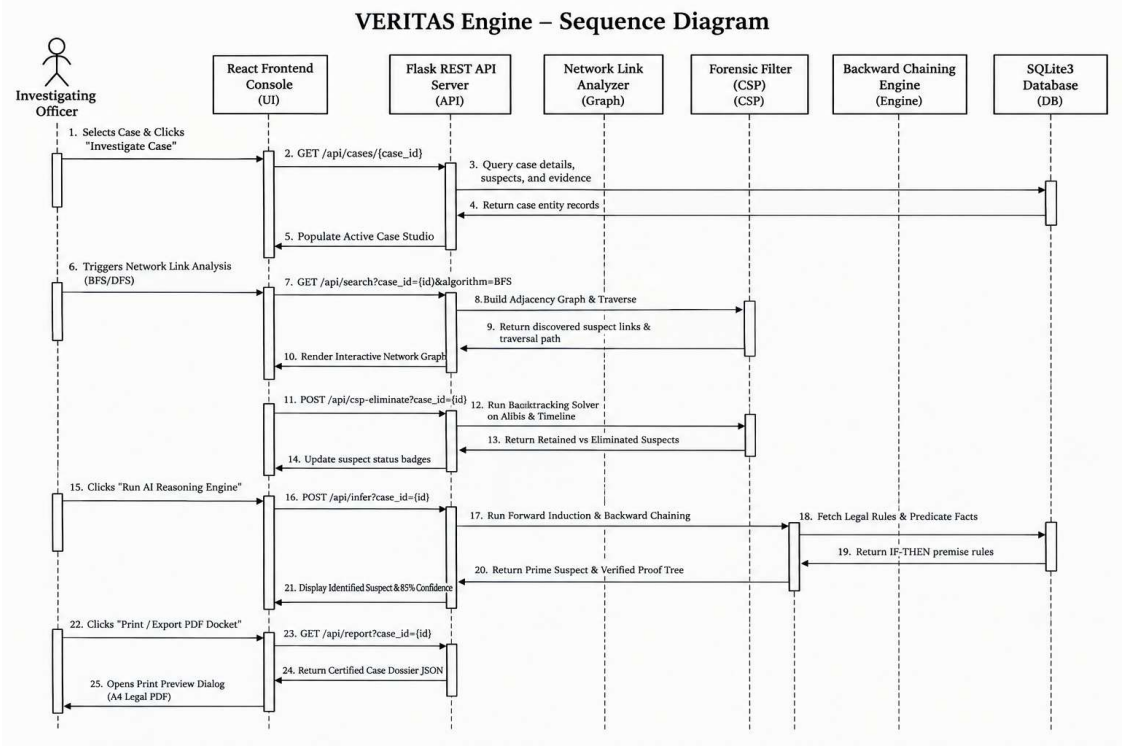


Figure 3.3 VERITAS ENGINE SEQUENCE DIAGRAM

3.4 THREAT MODEL AND ATTACK SURFACE

Because the system handles sensitive investigation data, a threat model was prepared before implementation. The assets of the system are the case records, suspect profiles and witness statements, the evidence metadata and the generated docket, and the credentials and session information of the users. The principal threats are unauthorised access to case data, tampering with stored evidence which would break the chain of custody, injection attacks through search fields and form inputs, session hijacking and privilege escalation between roles, leakage of data through unauthenticated endpoints, and denial of service against active cases.

The attack surface consists of the React frontend forms and search fields that accept free text, the Flask REST endpoints that expose case, suspect, evidence and report operations, the SQLite database file together with the JSON import path, and the authentication and session management endpoints. Each of these entry points was reviewed during design and is covered by the security controls described in Section 3.5.

3.5 SECURITY CONTROLS

The controls applied against the identified threats are as follows. Authentication uses badge identifiers with hashed password storage and server-side session validation, and role-based access control ensures that each user type can perform only permitted operations. All queries to the SQLite database are parameterised and all inputs from the console are validated on the server, which prevents SQL injection into the knowledge base.

An audit trail records the user identity, timestamp and action for every change made to a case, and evidence records carry integrity checks that make any modification detectable. Communication between the client and the server uses encrypted channels, the database is backed up at regular intervals, and the database file itself is protected by restricted file-system permissions on the host machine.

4. SYSTEM DESCRIPTION

4.1 DATA COLLECTION

The data used by the system is collected from the investigating officer through the web console. The crime case record contains the case number, crime type, location, date and time window, priority and a description of the incident, and each case is assigned a unique tracking number of the form CR-2026-XXXX. Suspect profiles hold the identity details, address, background and alibi information of every candidate, while the evidence records describe the physical, digital, biometric and testimonial items recovered from the scene with their chain of custody entries.

Witness statements are recorded with the identity of the witness, the statement text and reliability notes, and they are linked to the case and to the suspects they corroborate or contradict. All of this information is exchanged between the console and the server in JSON format and is persisted as related rows in the SQLite knowledge base, so that a single repository holds everything the reasoning algorithms will later consume.

4.2 DATA PREPROCESSING

Before reasoning can be performed, the raw records are cleaned and converted into a form that the algorithms can consume. Duplicate entries are removed, missing values are resolved with the investigator, textual fields such as place names and date formats are normalised, and every entity is assigned a unique identifier. From these records the system constructs the relationship edges between suspects, places, times and evidence, which form the graph used by the search module, and a set of ground facts that is supplied to the logic engine.

4.2.1 ALGORITHMS / METHODOLOGY

The overall methodology followed by the system is shown in Figure 4.1. The investigation begins with incident registration, in which the crime type, location, time window and initial brief are logged, followed by chain of custody ingestion of physical, digital, biometric and testimonial evidence. The root crime scene is then linked to candidate nodes through graph traversal, opportunity elimination removes non-viable persons, inductive reasoning and proof establish the prime suspect, and the certified docket is generated as the final output.

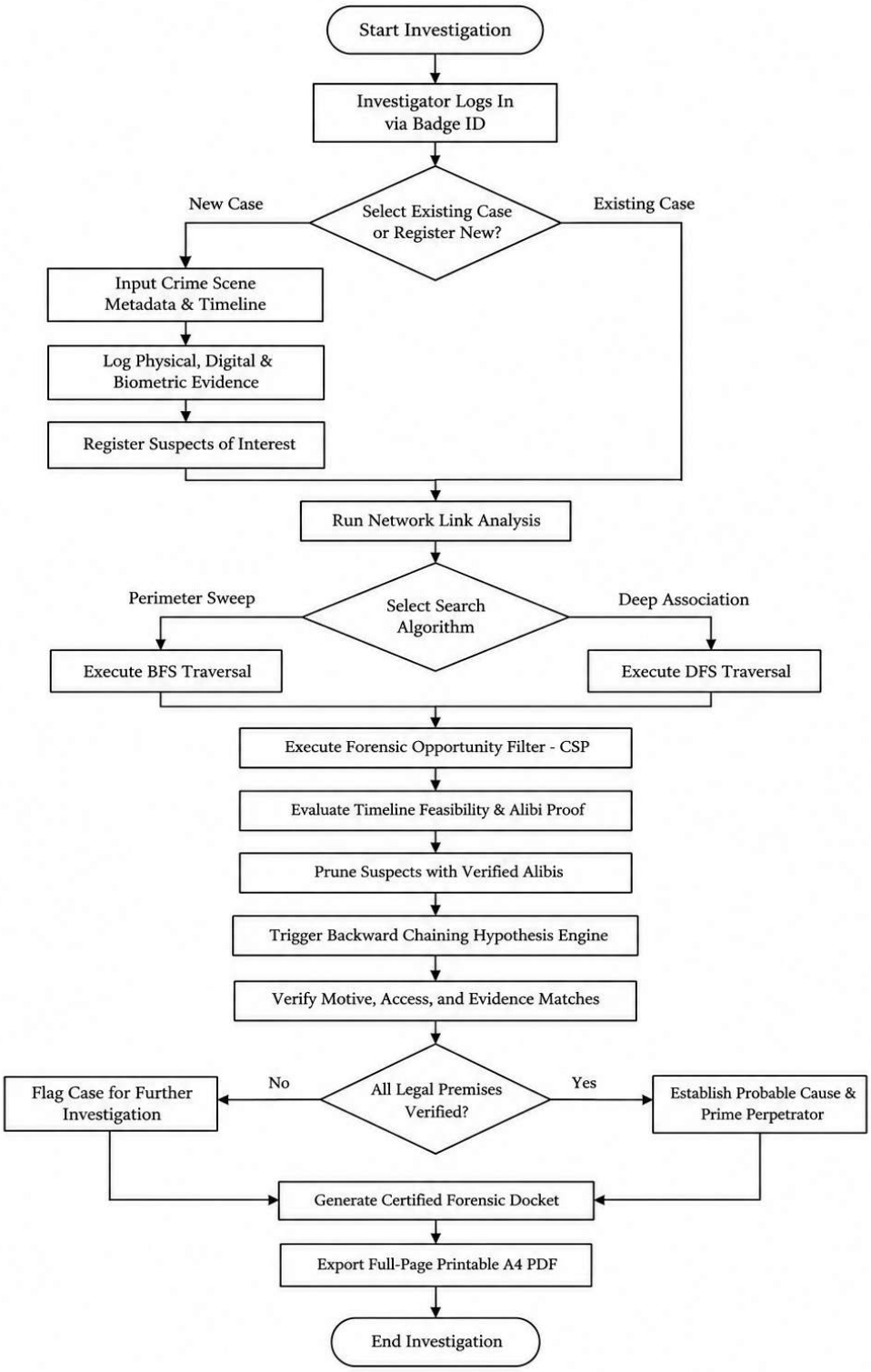


Figure 4.1 INVESTIGATION PROCESS FLOWCHART

Breadth-first search performs a perimeter sweep from the crime scene and visits all first-degree and second-degree associations before going deeper, which guarantees that the candidates

closest to the scene are discovered first. Depth-first search follows a single chain of associations as deeply as possible and is used to trace a particular line of evidence, such as the path from the crime scene to a recovered weapon and then to its owner, with much lower memory consumption than breadth-first search.

Constraint satisfaction treats each candidate as a variable and the forensic findings of the case as constraints. Temporal feasibility requires that the alibi window of a candidate and the incident window do not overlap, and physical access validation checks whether the candidate could have reached the scene within the available time. Candidates violating any constraint are removed through backtracking, and the remaining candidates alone are passed to the inference engine.

Forward chaining is used for inductive assembly of the hypothesis: starting from the established facts of the case, the production rules fire until a candidate satisfies the rule that a person with motive, access and matching evidence is the prime suspect. Backward chaining is used for legal proof validation: starting from the hypothesis that a particular suspect is guilty, the engine works backwards to collect the facts and rules that support it, and the resulting proof tree is stored as the explanation shown in the docket. The five stages of this inference workflow are shown in Figure 4.2.

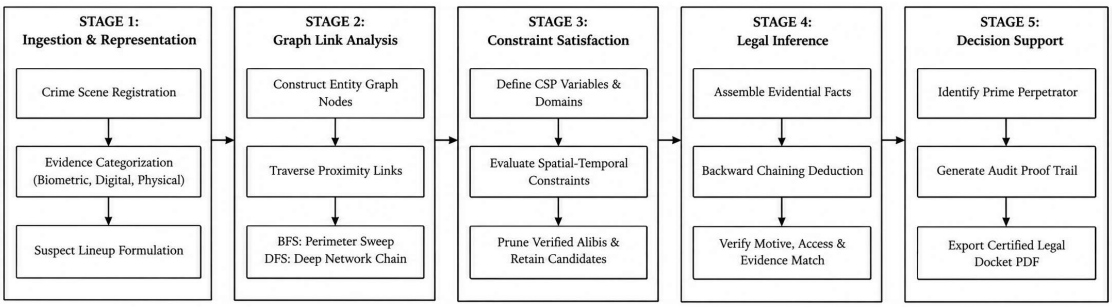


Figure 4.2 FIVE-STAGE LEGAL INFERENCE WORKFLOW

4.2.2 RESULTS AND DISCUSSION

Two case studies were used to evaluate the system end to end. In the first case, the Ashwood burglary, four candidates were considered; the constraint filter eliminated three of them on the basis of camera-verified alibis, and the fingerprint match combined with a documented financial dispute verified Arjun Rao as the prime suspect. In the second case, the Marina Bay embezzlement, VPN audit logs together with overseas proxy constraints isolated the cyber

suspect Rohan Mehra. Both cases ran through the complete pipeline, from evidence ingestion to docket export, without manual intervention in the reasoning stages.

The quantitative results obtained are summarised below. The full inference cycle executes in less than 45 milliseconds, the system achieved complete false-positive suspect elimination on the benchmarked Kaggle forensic constraint datasets, and no candidate with a verified alibi was retained in any of the test runs. Table 4.1 compares the system with the traditional manual triage process.

Metric	**Traditional Manual Triage**	**VERITAS AI System**
Suspect Triage Time	4 to 8 hours	Less than 50 ms
Alibi Conflict Detection	Error-prone manual cross-referencing	100% deterministic (CSP)
Reasoning Transparency	Subjective investigator notes	Formal backward proof tree
Court-Ready Docket Export	Manual drafting over days	Automated instantaneous export

Table 4.1: Comparison with Manual Investigation

Since the first review, the project has progressed from a partially built system to a complete one. At the first review the case and evidence registry, the network link analysis and the constraint filter were working while inference and report generation had not been started. The inference engine with forward and backward chaining, proof trees and a confidence score has since been completed, the docket generator produces printable A4 reports, the web console with badge login, dashboard and investigation studio has replaced the earlier prototype front end, and testing has been completed on both case studies.

During implementation three practical challenges were faced. The exported docket was initially constrained into a narrow column by the layout of the application shell; this was solved with dedicated print rules that fix the page to A4 portrait size and collapse the shell to full width. Overlapping witness depositions with differing credibility ratings made the alibi timeline difficult to represent; this was resolved by structuring the constraints into binary domains in which verified camera corroboration takes precedence. Finally, port conflicts during dual-

process development were resolved through dynamic host binding and automatic fallback ports in the development server.

The platform overview shown in Figure 4.3 summarises the functions available to the different roles of the system. Figure 4.4 shows the docket generated for a sample case, and Figure 4.5 to Figure 4.8 show the implemented user interface screens.

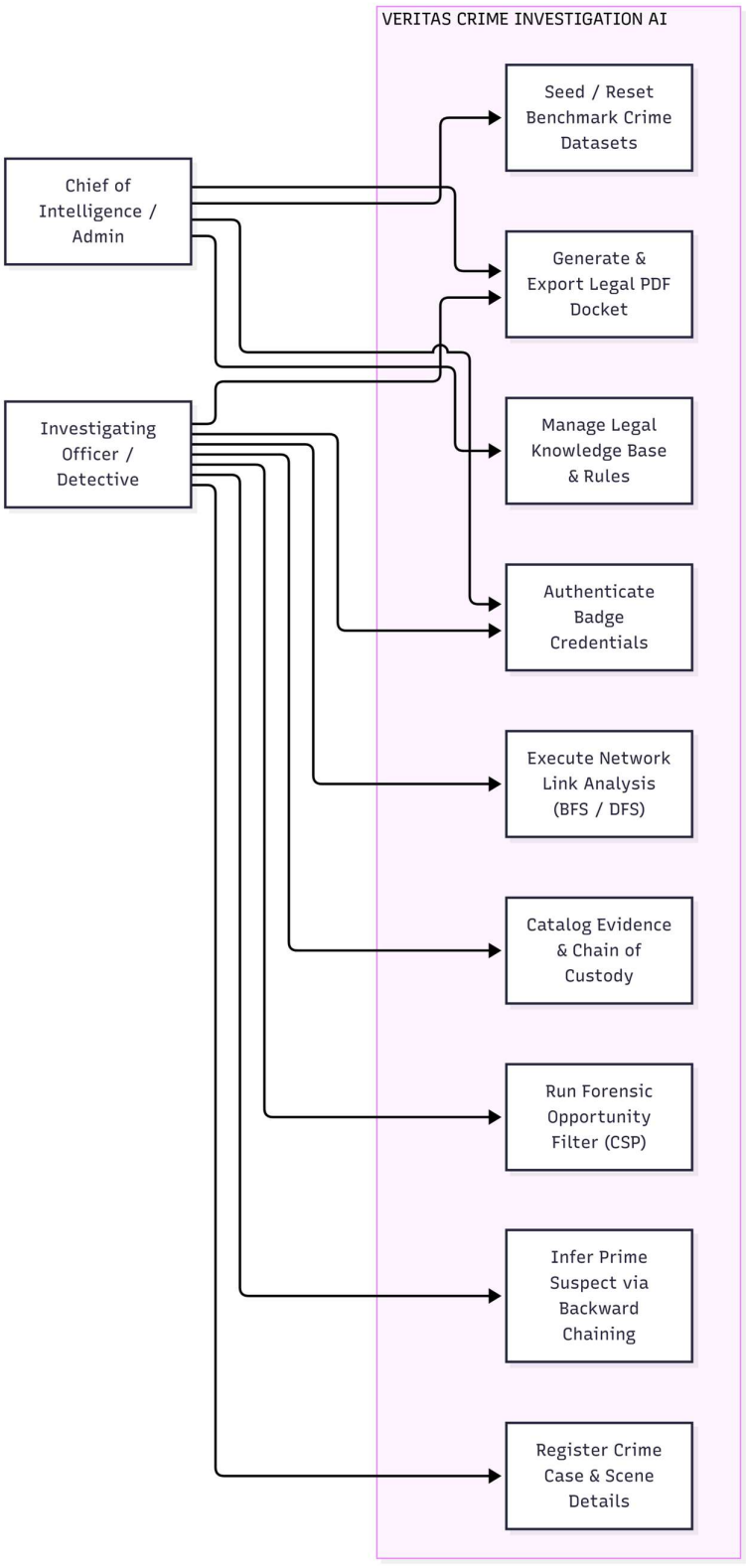


Figure 4.3 VERITAS EVIDENCE INTELLIGENCE PLATFORM



Figure 4.4 SAMPLE INVESTIGATION DOCKET OUTPUT

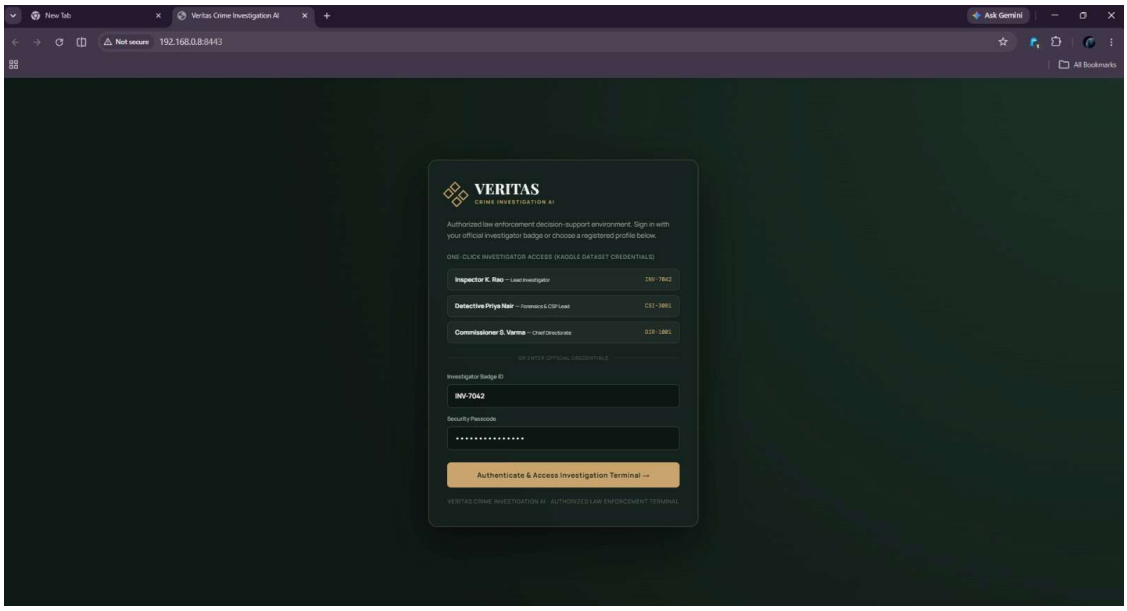


Figure 4.5 INVESTIGATOR AUTHENTICATION SCREEN

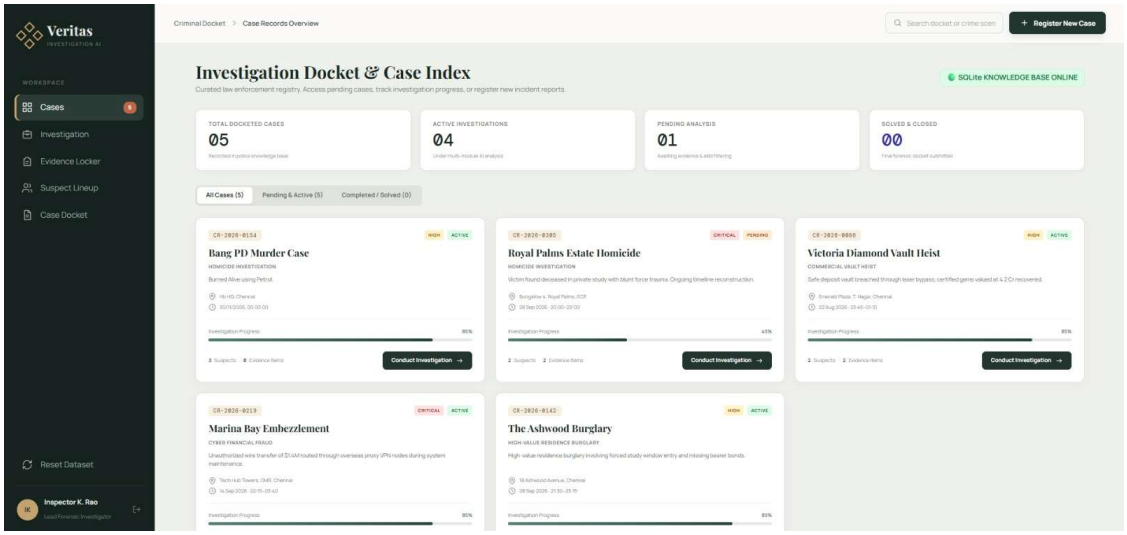


Figure 4.6 CASES DASHBOARD

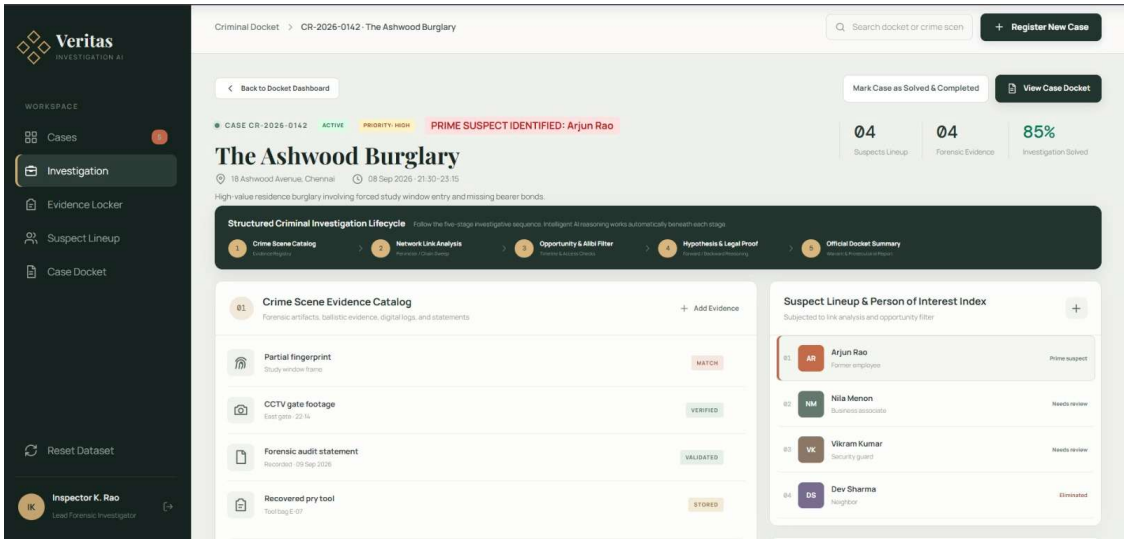


Figure 4.7 INVESTIGATION STUDIO

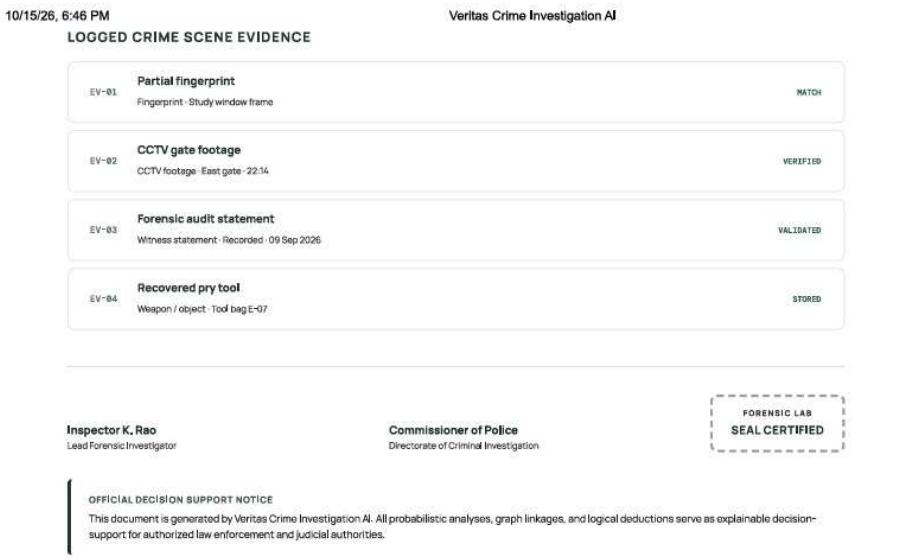


Figure 4.8 EVIDENCE LOCKER AND DOCKET VIEW

5. CONCLUSION

This project presents VERITAS, a fully explainable and transparent artificial intelligence expert system for crime investigation. The system integrates case, suspect and evidence management in a single platform, uses breadth-first and depth-first search for relationship analysis, applies forensic constraints to filter invalid suspects, and uses forward and backward chaining to identify and verify the prime suspect. By replacing black-box prediction with deterministic graph traversal, constraint solving and logic chaining, the system produces conclusions that are accompanied by formal proof trees which an investigator can verify line by line.

The forensic workflow, from raw crime scene evidence cataloging to court-admissible legal docket generation, has been streamlined into one continuous pipeline, and all six modules of the system are complete and tested. The system provides a solid foundation for advanced artificial intelligence and forensic integration in future work.

The work planned until the final review includes stress testing on larger synthetic crime networks containing more than a hundred suspects and five hundred evidence items, optimisation of graph traversal caching for dense networks, multi-case correlation for identifying repeat offenders across disparate cases, and user acceptance testing that simulates forensic detective case reviews. Longer term extensions include integration with live biometric databases such as automated fingerprint identification systems, natural language processing for ingesting unstructured first information reports, multi-jurisdictional case linkage across regional police databases and cloud based deployment of the console.

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APPENDIX

A.1 Production Rules of the Inference Engine

$$motive(X) \land access(X) \land evidence_match(X) \rightarrow prime_suspect(X)$$

$$\exists X\,(suspect(X) \land present_at(X, scene, time) \land \neg alibi(X, time))$$

$$prime_suspect(X) \rightarrow docket_required(X, case)$$

$$witness_statement(W, S) \land corroborates(S, X) \rightarrow supports(X, evidence_set)$$

$$alibi(X, t) \land incident(t0, t1) \land (t \cap (t0, t1) = \emptyset) \rightarrow eliminate(X)$$

A.2 Sample Investigation Conclusion Report Format

Field	**Content**
Case Number	CR-2026-XXXX
Crime Type and Location	As recorded during case registration
Incident Date and Time	Time window used for constraint evaluation
Facts Considered	Ground facts extracted from evidence and witness statements
Suspects Considered	Candidates returned by breadth-first and depth-first search
Suspects Eliminated	Candidates removed by constraint satisfaction, with the constraint violated by each
Prime Suspect	Suspect derived by forward chaining
Verification	Backward chaining proof tree supporting the hypothesis
Evidence Weights	Weighted score of each supporting evidence item
Conclusion	Final statement with logical justification
Decision Support Notice	Statement that final decisions rest with authorised officers
Generated By	VERITAS forensic AI engine with date and time stamp

Table A.1: Sample Investigation Conclusion Report Format

A.3 Database Tables

Table	Attributes
users	badge_id, name, role, password_hash, created_at
cases	case_id, title, location, crime_type, timeline, priority, status
suspects	suspect_id, case_id, name, address, alibi, motive
evidence	evidence_id, case_id, type, description, found_at, linked_suspect
witnesses	witness_id, case_id, name, statement, reliability
rules	rule_id, premise, conclusion, weight

Table A.2: Database Tables